

Ioan Lucut

Lead Engineer – Search and product platforms, from architecture to production

Remote, EU | [linkedin.com/in/ilucut](https://www.linkedin.com/in/ilucut) | github.com/ioanlucut | ioan.lucut88@gmail.com

Professional Summary

Hands-on Lead Engineer who led ImmoScout24 Austria's property-search Portal architecture and kept evolving it through search, performance, privacy and AI. Builds and operates customer platforms and payment workflows.

Work Experience

ImmoScout24 Austria | Vienna

Mar 2016 – Present

Current role: Lead Engineer, since 2018 | Remote

- Shape technical direction, priorities and programme staffing with fellow leads and the Head of Technology; stay hands-on in implementation and production support.
- Interview and help select engineers, for about nine years; mentor colleagues and onboard engineers to development and release practices.

Portal architecture and search | 2018 – Present

- Led the 2018–2019 **Portal relaunch architecture**, implementing streaming React SSR, the initial GraphQL search API and Elasticsearch-backed search URLs. The platform **remains in production** in 2026.
- Led the **legacy catalog migration** into the Portal in 2019–2020, with phased redirects, tested SEO compatibility and coordinated cutovers. The SEO team reported higher contact-mail conversion on the new result list than on the old catalog (April 2020).
- Led **map search** from feasibility to first production release in three months (2025). Designed and built Elasticsearch aggregation around sampled real property locations rather than cluster centroids; production aggregation **p95 was under 100 ms**.
- Designed and built **stable location masking** after customers reported hidden-address listings appearing at their true location (2026). Masked pins stay within postal-code polygons; rollout included reimport checks and fallback monitoring.
- Built and released the **grid and mini-map result page** in August 2026, using screen-resolution analytics to shape the layout and shipping a backward-compatible GraphQL API ahead of the frontend.
- Delivered **official-region search** across APIs, Portal URLs, saved searches and the sitemap; rebuilt geographic autocomplete with context-based area selection (2021–2022).
- Built the Portal integration for **AI image-tag search** (2025), including GraphQL and fuzzy autocomplete, so seekers can find visual features realtors never tagged. The AI team owned image extraction.
- Led Austrian delivery of an **embedding-based AI recommender**, structuring colleagues' implementation work, defining go-live conditions and designing the staged app-feed rollout. The recommender went live in the Austrian apps in September 2026.
- Supplied the recommender's training-data contract, identity stitching and event pipelines. The German AI Solutions team, coordinated by a senior ML engineer, owned the model and API, using SageMaker inference and OpenSearch vector search.
- Started the **web-performance and SEO programme** in 2021: fixed layout shift, built Chrome UX Report monitoring and drove crawl-efficiency work. Googlebot's share of requests to JSON endpoints fell **from 75% to 48%** between November 2021 and January 2022.

- Led the later **INP/Core Web Vitals work**, reworking Tealium's tag loader to yield between tasks and directing follow-ups. The SEO team's reports recorded the site under its INP goal from January 2025 and **all URLs passing Core Web Vitals** by mid-2026.
- Wrote the technical concept and ADR for **vacation-property search** and created the team's implementation stories; built the export of Austrian realtors' foreign listings to ImmoScout24 Germany; extended the country coverage after the December 2023 team launch.
- Led the Portal side of **virtual-viewing search** in the March 2020 lockdown: filter demoable two days after it began, **360°/video tours live on day eight**. By 1 April, ImmoScout24 reported over 15,000 listings with video viewings.

Customer platforms and production workflows | 2018 – Present

- Architected and initially built **My Profile** for realtor and private-customer self-service (2018). Later led the pay-per-listing frontend replacement for private sellers and realtors, bringing editing, product selection, verification, Adyen payment and publication into React (2020–2021).
- Built **realtor slot self-service** for assigning marketing placements from existing contracts. Support reported **33% fewer tickets** in May–June 2021 compared with the same period in 2020.
- Built the **Salesforce payment-method API** and coordinated the November 2024 Salesforce/Adyen cutover of **17,000+ accounts** without notable issues. Reusable SEPA and card methods let listings publish before payment collection.
- Replaced fragile **payment-completion sequences** with Step Functions and Lambda from December 2024, making failures visible, recoverable and easier to debug. Added end-to-end tests for retries, failure paths and completion.
- Built **retryable Zuora billing callbacks** in March 2025, starting with invoice fallback after failed renewals, with dead-letter recovery and a guard bounding the consumer's daily Salesforce API calls (OAuth-protected API Gateway, SQS, Lambda, DynamoDB).
- Led the **AI-assisted fraud-review integration** with Customer Service and another engineer (2026). Built the Dify/Claude-on-Bedrock path into private-listing publishing, with versioned workflows, manual fallback, audit records and a Support-operated kill switch.
- The workflow automated first-pass checks that Support had made by hand, weekends included. In June–August 2026, Support reported **half of private-listing checks** clearing for immediate publication instead of manual review.
- Rebuilt the **mortgage financing calculator** as its main engineer (2021), adding an affordability check and listing prefill. The tool recorded **1,300+ lead emails** in the 30 days after its November update.

Engineering foundations and modernisation | 2016 – Present

- Co-created and maintain the **shared React components** and build tooling underpinning the Portal and other Austrian frontends, alongside continued application and framework modernisation.
- Rewrote **immodirekt.at** and **immobilien.net** on Node.js and React as their main developer, with server-side rendering and search-friendly result-list URLs. The immodirekt.at rewrite remains **in production nine years later**.
- Turned a hackathon prototype into **per-pull-request preview environments** for the listing-detail application in 2023, with automatic cleanup. Still used by colleagues in 2026, each environment lets them deploy and test independent changes in parallel rather than queue for one shared sandbox.
- Built the **ID5 advertising-identity integration** as tested code (2025). Designed a queue to hold early tracking events until the asynchronous ID arrives, then replay them enriched; after two seconds, events are released without it rather than remaining blocked.
- With a colleague, traced the listing-detail service's **production memory leak** through load tests and heap dumps (2022) and shipped the fixes; then reduced minimum reserved **CPU by 55% and memory by 60%**.
- Lead the team's **AI-native engineering initiative** (2026); wrote the plan for agent-ready repositories, a cross-service context layer and safe automation of recurring work.

- Designed and rebuilt the responsive property-search map on Solr: the API chose the cluster level from zoom and result count, zooming refetched clusters and list, and heuristics linked list and pins when exact coordinates were missing.
- For the Hummingbird redesign of search and detail pages, rewrote the JSP result list (including Top Properties) and listing-detail page with custom tags, a new gallery and price display, without regressions. Built the booking back end customers used to place their ads in the Pole Position ranking (2014).
- Extended SEO-friendly URLs from a few main page types to the real-estate and motor categories (desktop, API, mobile web); built an Apache Camel job generating interlinking URLs.
- Built the Solr search-result display in both front ends (Spring desktop, AngularJS mobile web); on mobile web, launched quick search with autocomplete and hid site navigation in the native apps.

Software Engineer | DiTech GmbH | Vienna

Jun 2012 – Apr 2014

- Co-built the live mobile web shop in a team of two, from the design team's specs: product search, filters and product pages (a colleague built most of checkout). Built a B2B "Business Class" section of the Java/JSP desktop shop.
- Built ERP test automation: a Selenium Grid suite, load runs and multi-threaded database tests that exposed a deadlock and stock-calculation errors; built the ERP's change-history (audit trail) module.
- Built DiBid, a supplier reverse-auction platform linked to the ERP: buyers opened auctions from purchase orders and chose winning bids (Java EE, JSF, PostgreSQL). It was complete when the company closed, so it never launched.

Software Engineer | iQuest Group | Cluj-Napoca

Feb 2012 – May 2012

- Built features for Virgin Atlantic's Java/Spring flight-booking platform with UK stakeholders.

Automation Engineer | Emerson | Cluj-Napoca

Sep 2010 – Jan 2012

- Developed control logic for the Emile Huchet power-plant modernisation; participated in factory acceptance testing in Lyon and on-site commissioning.

Skills

Languages and frameworks: TypeScript, JavaScript, React, Node.js, GraphQL, Elasticsearch, server-side rendering, CSS Modules; Java, Spring.

Architecture and cloud: System design, serverless and event-driven architecture, infrastructure as code; AWS Lambda, Step Functions, API Gateway, SQS, DynamoDB, ECS, CloudFormation, S3, CloudFront; Docker, Datadog.

Quality and delivery: Jest, Playwright, Storybook, end-to-end testing, Jenkins CI/CD, code review, web performance and SEO.

AI and business systems: Claude on Amazon Bedrock, Dify, ML recommender API integration, AI-assisted engineering with Claude Code and Codex; Adyen, Zuora, Salesforce, Tealium.

Education

MSc, Advanced Process Control – Technical University of Cluj-Napoca | 2010 – 2014

BSc, Automation and Computer Science – Technical University of Cluj-Napoca | 2006 – 2010

Languages: Romanian: Native | English: Fluent | German: C1 | French: Basic